

Minimal Notes

Real Analysis

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WEEK 1 - LIMITS

Sets

Definition 1 (Types of sets). Suppose that $A \subset \mathbb{R}$ then we say A is

- **Bounded** if it can be contained in an interval $[m, n]$ for some $m, n \in \mathbb{R}$.
- **Open** if at any $x \in A$ we can fit an interval (a, b) in A such that $x \in (a, b) \subset A$.
- **Closed** if $\mathbb{R} \setminus A$ is open.
- **Countable** if there is injection $A \rightarrow \mathbb{N}$. (i.e. either finite or of the same cardinality as $\mathbb{N}$.)

Ex1 q3

Sequences

Definition 2. • **Convergence** $a_n \rightarrow a$: For any $\varepsilon > 0$ there is $N \in \mathbb{N}$ so that for all $n \geq N$:

$$|a_n - a| < \varepsilon.$$

- **Cauchy**: For any $\varepsilon > 0$ there is $N \in \mathbb{N}$ so that for all $n, m \geq N$:

$$|a_n - a_m| < \varepsilon.$$

Recall that a sequence converges if and only if it is Cauchy.

Theorem 1 (Bolzano-Weirstrass). A bounded sequence has a convergent subsequence.

Limits of Functions

Definition 3. • **Limit of a function at a point**: Let $x_0 \in (a, b)$ and take $f : (a, b) \setminus \{x_0\} \rightarrow \mathbb{R}$. Ex1 q5

We write $\lim_{x \rightarrow x_0} f(x) = y_0$ if:

For any $\varepsilon > 0$ there is $\delta > 0$ so that for any x with $0 < |x - x_0| < \delta$ we have $|f(x) - y_0| < \varepsilon$.

- **Infinite limit of a function**: Let $x_0 \in (a, b)$ and take $f : (a, b) \setminus \{x_0\} \rightarrow \mathbb{R}$. We write $\lim_{x \rightarrow x_0} f(x) = \infty$ if:

For any $M > 0$ there is $\delta > 0$ so that for any x with $0 < |x - x_0| < \delta$ we have $f(x) \geq M$.

- **Limit a function at infinity**: let $f : (a, \infty) \rightarrow \mathbb{R}$. We write $\lim_{x \rightarrow \infty} f(x) = y_0$ if:

For any $\varepsilon > 0$ there is $R > a$ so that for any $x \geq R$ we have $|f(x) - y_0| < \varepsilon$.

- **Left(-)/Right(+) limit**: let $f : (a, b) \rightarrow \mathbb{R}$. We write $\lim_{x \rightarrow x_0^+} f(x) = y_0$ if:

For any $\varepsilon > 0$ there is $\delta > 0$ so that for any x with $x_0 < x < x_0 + \delta$ we have $|f(x) - y_0| < \varepsilon$.

Note that in particular the function does not need to be equal to the limit i.e. $f(x_0) = \lim_{x \rightarrow x_0} f(x)$ or even be defined at x_0 for the limit to exist at x_0 .

WEEK 2 - CONTINUITY

Definition 4 (Continuity at a point). Let $f : (a, b) \rightarrow \mathbb{R}$ and $x_0 \in (a, b)$. We say f is **continuous at** x_0 if Ex2 q3

For any $\varepsilon > 0$ there is $\delta > 0$ so that for any x with $|x - x_0| < \delta$ we have $|f(x) - f(x_0)| < \varepsilon$.

A function is continuous on an interval if it is continuous at any of its points. The set of all continuous function on $[a, b]$ is denoted $C^0[a, b]$ ¹ (or just $C[a, b]$).

Note that the only difference between the definitions of continuity and a limit at x_0 is the " $0 <$ " part. Namely continuity requires the function to be defined at $f(x_0)$. The definitions become equivalent if the limit is $f(x_0)$ in which case f is defined at x_0 :

Theorem 2. • $\lim_{x \rightarrow x_0} f(x) = y_0$ if and only if

$$\lim_{x \rightarrow x_0^+} f(x) = y_0 = \lim_{x \rightarrow x_0^-} f(x).$$

• (cts iff left and right limits agree) f is continuous at x_0 if and only if

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0) = \lim_{x \rightarrow x_0^-} f(x)$$

if and only if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

Definition 5 (Types of discontinuities). • **Removable:** If $\lim_{x \rightarrow x_0^+} f(x) = y_0 = \lim_{x \rightarrow x_0^-} f(x)$ but $f(x_0) \neq y_0$. (Eg. $f = x/|x|$ at $x = 0$) Ex2 q5

• **Jump:** If $\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x)$ exist but are not equal. (Eg. $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$ at $x = 0$)

• **Infinite:** If $\lim_{x \rightarrow x_0^+} f(x)$ or $\lim_{x \rightarrow x_0^-} f(x)$ is infinite. (Eg. $f = 1/x$ at $x = 0$)

• **Oscillatory:** If $\lim_{x \rightarrow x_0^+} f(x)$ or $\lim_{x \rightarrow x_0^-} f(x)$ does not exist. (Eg. $f = \sin(1/x)$ at $x = 0$)

Theorem 3 (Continuity via sequences). Let $f : (a, b) \rightarrow \mathbb{R}$ and $x_0 \in (a, b)$. Then

f is continuous at x_0 if and only if for any sequence² $x_n \rightarrow x_0$ we have

$$\lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) = f(x_0).$$

In other words continuous functions allow us to pass limits on the inside. **Exam Tip:** The theorem is also very useful when trying to show a function is *not* continuous at x_0 : It suffices to produce a sequence $x_n \rightarrow x_0$ such that $\lim_{n \rightarrow \infty} f(x_n) \neq f(x_0)$.

Algebra of Continuous functions: sums, products, fractions and compositions are continuous under the standard conditions. Ex2 q6,7

¹Endpoints understood as one-sided limits.

² x_n are in $(a, b) \setminus \{x_0\}$.

WEEK 3 - CONTINUITY

Basic theorems on Continuity

Theorem 4. • (*Boundedness theorem*) A continuous function on a closed bounded interval is bounded.

• (*Maximum, Minimum theorem*) A continuous function on a closed bounded interval achieves its maximum and minimum.

• (*Intermediate Value theorem*) A continuous function on $[a, b]$ attains all values between $f(a)$ and $f(b)$. Ex3
q2,3,5

The boundedness theorem can be extended to an unbounded interval: If $f : [a, \infty) \rightarrow \mathbb{R}$ is continuous and $\lim_{x \rightarrow \infty} f(x)$ exists then f is bounded on $[a, \infty)$.

Uniform Continuity

Notice that the δ in **Definition 4** of pointwise continuity depends on the point x_0 and on the chosen ε . This means at two different points x_1 and x_2 we may have to choose different $\delta(\varepsilon, x_1)$ and $\delta(\varepsilon, x_2)$ to remain within ε of $f(x_1)$ resp. $f(x_2)$. Uniform continuity does *not* depend on the choice of the point:

Definition 6. Let I be an interval. The function $f : I \rightarrow \mathbb{R}$ is **uniformly continuous** if

For any $\varepsilon > 0$ there is $\delta > 0$ so that for any x, y with $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$.

Exam Tip: To show a function is *not* uniformly continuous it suffice to find sequences x_n, x'_n so that $\lim_{n \rightarrow \infty} |x_n - x'_n| = 0$ and yet $\lim_{n \rightarrow \infty} |f(x_n) - f(x'_n)| \neq 0$.

Uniform continuity always implies pointwise continuity but the reverse implication occurs only if I is closed and bounded:

Theorem 5 (Cantor's theorem). If f is continuous on a closed and bounded interval I then f is uniformly continuous on I .

WEEK 4 - DERIVATIVES

The focus of this week is the study of fractions of the form

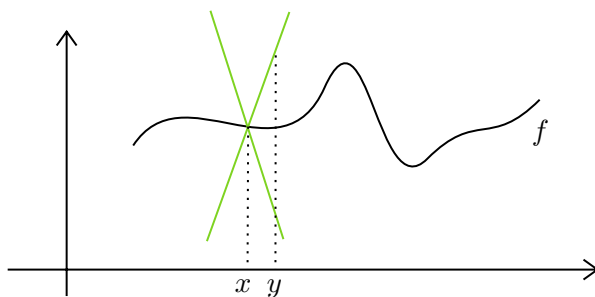
$$\frac{f(x)-f(y)}{x-y} \text{ for some function } f : I \rightarrow \mathbb{R}.$$

The first use case of the fraction is a type of continuity.

Definition 7 (Lipschitz continuity). A function $f : I \rightarrow \mathbb{R}$ is **Lipschitz continuous** if there is some $C > 0$ so that

$$|f(x) - f(y)| \leq C|x - y| \text{ for any } x, y \in I.$$

Intuition: The ratio $\frac{f(x)-f(y)}{x-y}$ measures the slope of f between two points x, y . By including the absolute values and letting y vary while x is fixed, we see that for f to be Lipschitz it needs to fit within a cone of slope C centered at x as it varies.



The next use of the fraction is the definition of a derivative.

Definition 8 (Derivative). *The function $f : I \rightarrow \mathbb{R}$ is differentiable at $x_0 \in I$ if the following limit exists*

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{equivalently} \quad \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

*If it exists, it is denoted $f'(x_0)$. If the k -th derivative exists and is continuous we say f is **k -times continuously differentiable** and denote the class of such functions $C^k(I)$.*

Exam Tip: The derivative limit exists if and only if the right and left limits exist.

Ex4 q2

Example 1 (Function that is once but not twice differentiable): Define $f(x) = 0$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$ (Draw this). Then computing the left and right limits at $x = 0$:

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0 - 0}{h} = 0 \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{(0+h)^2 - 0^2}{h} = 0.$$

Since the limits agree $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = f'(0)$ exists and equals 0. Elsewhere we use rules of differentiation to see that

$$f'(x) = \begin{cases} 0 & x < 0 \\ x & x \geq 0. \end{cases}$$

Computing the left and right derivatives of $f'(x)$ we obtain

$$\lim_{h \rightarrow 0^-} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0 - 0}{h} = 0 \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1.$$

Since the left and right limits do *not* agree it follows the derivative does not exist so the function f is not twice differentiable.

Algebra of Differentiable functions: sums, products, fractions and compositions are differentiable Ex4 q5 under the standard conditions.

The following theorem says every differentiable function is continuous. The reverse is clearly not true (e.g. $|x|$ is continuous but not differentiable at 0).

Theorem 6.

$$\{\text{Differentiable on } I\} \subset \{\text{Cts on } I\}.$$

Mean Value Theorem

We say $c \in I$ is a **local maximum** of f if for some $\varepsilon > 0$ we have $f(c) \geq f(x)$ for all $x \in (c - \varepsilon, c + \varepsilon)$.³

Theorem 7. • (**Fermat's theorem**) If f is differentiable on (a, b) and has local maxima at $c \in (a, b)$, then $f'(c) = 0$.

• (**Rolle's theorem**) Suppose f is continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$ then there is $c \in (a, b)$ with $f'(c) = 0$.

• (**Mean Value theorem**) Suppose f is continuous on $[a, b]$ and differentiable on (a, b) . Then there is $c \in (a, b)$ with Ex4 q6

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

As an application of some of the above we prove relative containment of some of the classes of functions we've met.

Theorem 8.

$$\{C^1(I) \text{ with bounded derivative}\} \subset \{\text{Lipschitz on } I\} \subset \{\text{Unif. cts on } I\} \subset \{\text{Cts on } I\}$$

Proof. For the first containment we use MVT. For the second it suffices to take $\delta < \varepsilon/C$ in the definition of Lipschitzness. The last follows by definition. $\square$

Exam Tip: The containment is a particularly fast way of showing a function is uniformly or Lipschitz continuous. Essentially, it says that if the derivative is bounded, then the function is Lipschitz and so uniformly continuous. (To show that the derivative is bounded use the Boundedness theorem.) Ex3 q6,7

Taylor Series

One way to think of MVT is that it gives us a **linear approximation** of f near x_0 using the first derivative:

$$f(x) = f(x_0) + f'(c)(x - x_0) \quad \text{for some } c \text{ between } x \text{ and } x_0.$$

If we also know the second derivative we can give **quadratic approximation**:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2.$$

If we know n derivatives we can approximate f by a polynomial with n -th degree polynomial⁴:

Theorem 9 (Taylor's formula). Suppose $f \in C^n(a, b)$ and $x_0 \in (a, b)$. Then for any $x \in (a, b)$ there is c between x and x_0 with Ex4 q7

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n)}(c)}{n!} (x - x_0)^n.$$

Note that $\frac{f^{(n)}(c)}{n!} (x - x_0)^n = O(|x - x_0|^n)$ as $x \rightarrow x_0$.

³Similarly for local minimum.

⁴The higher the degree of the approximating polynomial the better the approximation is. Knowing information about more derivatives gives us more information about the function.

WEEK 5 - ANALYSIS OF FUNCTIONS

A function is increasing if for $x \leq y$ we have $f(x) \leq f(y)$. It is strictly increasing if $f(x) < f(y)$.

Theorem 10 (Monotonicity using derivatives). *Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable.*

Ex5 q1

- If $f'(x) \geq 0$ (resp. ≤ 0) for all $x \in (a, b)$ then f is increasing (resp. decreasing). If $f'(x) > 0$ (resp. < 0) then f is strictly increasing (resp. decreasing).
- (Zero derivative implies constant) If $f'(x) = 0$ for all $x \in (a, b)$ then f is constant.
- If $f'(x) = g'(x)$ for all $x \in (a, b)$ then $f = g + c$ for some constant $c \in \mathbb{R}$.

Recall that derivatives give us a way to find local extrema:

Theorem 11 (Local Extrema using derivatives). *Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable.*

- If $f'(c) = 0$ and $f''(c) > 0$ (resp. < 0) then c is a local minima (resp. maxima).

Convexity

A function $f : (a, b) \rightarrow \mathbb{R}$ is **convex** if for all $x, y \in (a, b)$ and any $\lambda \in [0, 1]$ we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

If the reverse inequality holds then the function is **concave**. Informally, convex functions look like $\cup$ and concave like $\cap$. Instead of laboriously checking convexity using the definition we can use the second derivative.

Theorem 12 (Convexity using second derivative).

- $f''(x) \geq 0$ (resp. ≤ 0) for all $x \in (a, b)$ if and only if f is convex (resp. concave).

Recipe for Global Extrema: Find global extrema of $f : [a, b] \rightarrow \mathbb{R}$.

Ex5 q5

- Find all points $c_1, \dots, c_n$ where $f'(c_i) = 0$.
- Evaluate $f(a), f(b), f(c_1), \dots, f(c_n)$ and find the maximum value. This is the global extremum. If the function is of the form $|f|$ then see if there are points with $|f| = 0$ and if yes include them in the set under consideration (see Ex5 q5b).

Recipe for Drawing Functions: Sketch f .

Ex5 q6

- Determine **Domain**.
- **Limits** at boundary and asymptotes (recall $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + O(x^3)$ for $|x| < 1$).
- Find and classify **discontinuities** (removable, jump, infinite, oscillatory).
- Find **zeros** and intervals of the same sign.
- Find local **extrema** ($f'(x) = 0$).
- Find intervals of monotonicity. **Tip:** Typically not needed so do not do unless specifically asked.
- Find intervals of convexity. **Tip:** Typically not needed so do not do unless specifically asked.

WEEK 6 - RIEMANN INTEGRATION

The upper (resp. lower) rectangle of a function f on some interval $[z_1, z_2]$ is the rectangle with base $[z_1, z_2]$ and height the largest (resp. lowest) value f takes on $[z_1, z_2]$ which is $\sup_{x \in [z_1, z_2]} f(x)$. The upper (resp. lower) sum for some partition $[a, z_1], [z_1, z_2], \dots, [z_{n-1}, b]$ is the sum of all the corresponding upper (resp. lower) rectangles:

Definition 9 (Upper and lower sums). *Let $f : [a, b] \rightarrow \mathbb{R}$. Then for a partition P with $a = z_0, \dots, z_n = b$ define the upper and lower sums as*

$$U(f, P) = \sum_{i=0}^{n-1} |z_{i+1} - z_i| \sup_{x \in [z_i, z_{i+1}]} f(x) \quad \text{and} \quad L(f, P) = \sum_{i=0}^{n-1} |z_{i+1} - z_i| \inf_{x \in [z_i, z_{i+1}]} f(x)$$

We also define

$$U(f) = \inf_P U(f, P) \quad \text{and} \quad L(f) = \sup_P L(f, P)$$

Definition 10 (Riemann Integrability). *A function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann Integrable **RI** if $U(f) = L(f)$.*

We need an easier criterion than the definition for determining whether a function is Riemann integrable. The most general is the following

Theorem 13 (Riemann Criterion). *A function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann Integrable if and only if for all $\varepsilon > 0$ there is a partition P so that*

$$U(f, P) - L(f, P) \leq \varepsilon.$$

The following theorem lists other easy to check sufficient conditions for Riemann Integrability:

Theorem 14 (Sufficient conditions for Riemann Integrability). *A function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann Integrable if at least one of the following holds:*

- f is monotonic.
- f is continuous.

Example 2 (Function that is not Riemann Integrable): Define $f(x) = 1$ if x is rational on $[0, 1]$ and $f(x) = 0$ otherwise. Then for any partition $U(f, P) = 1$ because any interval contains a rational so any upper rectangle has height 1. Similarly $L(f, P) = 0$ because any interval contains an irrational so the lower rectangle has height 0. Since this is true for any partitions it follows $U(f) = 1$ and $L(f) = 0$.

WEEK 7 - INTEGRATION 2

Theorem 15. • (Pieceswise composition) *If $g_1 : [a, b] \rightarrow \mathbb{R}$ and $g_2 : [b, c] \rightarrow \mathbb{R}$ are RI then so is f defined piecewise as $f = g_1$ on $[a, b]$ and $f = g_2$ on $[b, c]$.*

- (Lipschitz composition) *If $f : [a, b] \rightarrow \mathbb{R}$ is RI and $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is Lipschitz then $\phi \circ f$ is RI.*

Lemma 1 (Properties of an Integral). • $\int_a^b f = \int_a^c f + \int_c^b f$.

- (Linearity) $\int cf + g = c \int f + \int g$.
- (Monotonicity) If $f \leq g$ on $[a, b]$ then $\int_a^b f \leq \int_a^b g$.
- (Triangle inequality) $|\int f| \leq \int |f|$.

Ex7 q3,4

Theorem 16 (Jensen's inequality). If $f : [a, b] \rightarrow \mathbb{R}$ is RI and ϕ is convex on $f([a, b])$ then $\phi \circ f$ is RI and

$$\phi\left(\frac{1}{b-a} \int_a^b f(x) dx\right) \leq \frac{1}{b-a} \int_a^b \phi(f(x)) dx.$$

If ϕ was concave then the reverse inequality would hold.

The following theorem is central to analysis. It makes rigorous the intuitive notion that integration and differentiation are inverses of each other.

Theorem 17 (FTC). • (Derivative of an Integral) Let $f : [a, b] \rightarrow \mathbb{R}$ be RI then

$$F(x) = \int_a^x f(x) dx$$

is continuous on $[a, b]$. Additionally if f is continuous at x_0 then $F'(x_0) = f(x_0)$.

- (Integral of a Derivative) Have $F : [a, b] \rightarrow \mathbb{R}$. If
 - F continuous on $[a, b]$
 - F differentiable on (a, b) with $F' = f$
 - $f : [a, b] \rightarrow \mathbb{R}$ is RI.

Then

$$\int_a^b F'(x) dx = \int_a^b f(x) dx = F(b) - F(a)$$

In particular if $F \in C^1$ then $F' = f$ is continuous and so integrable and the result holds.

WEEK 8 - INTEGRATION 3

Theorem 18. • (By Parts) If $f, g \in C^1[a, b]$ then

$$\int_a^b f'(x)g(x) dx = f(x)g(x)|_a^b - \int_a^b f(x)g'(x) dx.$$

- (Substitution) If $f \in C^0[a, b]$ and $\phi \in C^1[a, b]$ then

$$\int_{\phi(a)}^{\phi(b)} f(x) dx = \int_a^b f(\phi(t))\phi'(t) dt.$$

Improper Integrals

We define

$$\int_{-\infty}^{\infty} f(x) dx := \lim_{b \rightarrow \infty} \int_{-b}^0 f + \lim_{b \rightarrow \infty} \int_0^b f \quad \text{and} \quad \int_a^{\infty} f := \lim_{b \rightarrow \infty} \int_a^b f.$$

If there is singularity at $c \in [a, b]$ then for $f : [a, b] \rightarrow \mathbb{R}$ we define

$$\int_a^b f(x) dx := \lim_{\varepsilon \rightarrow 0} \int_a^{c-\varepsilon} f + \lim_{\varepsilon \rightarrow 0} \int_{c+\varepsilon}^b f.$$

Example 3: • $\int_0^1 \frac{1}{x^\alpha} dx < \infty$ iff $\alpha < 1$.

• $\int_1^{\infty} \frac{1}{x^\alpha} dx < \infty$ iff $\alpha > 1$.

Recall if $\int_a^{\infty} |f| < \infty$ we say the integral converges **absolutely**. If only $\int_a^{\infty} f < \infty$ then the integral converges **conditionally**. By the triangle inequality absolute convergence implies conditional convergence. The reverse is not generally true.

Theorem 19. Let $f, g : [a, \infty) \rightarrow \mathbb{R}$. Assume f, g are RI on $[a, b]$ for any $b > a$.

• (Comparison test for Integrals) Assume $|f(x)| \leq g(x)$ for all $x \geq a$

Ex8 q4

$$\text{If } \int_a^{\infty} g < \infty \quad \text{then} \quad \int_a^{\infty} f < \infty.$$

In particular if $\int_a^{\infty} |f| < \infty$ then $\int_a^{\infty} f < \infty$.

• (Product theorem) Suppose the function $F(x) := \int_a^x f$ is bounded on $[a, \infty)$. Suppose that $g \in C^1[a, \infty)$, monotone and $\lim_{x \rightarrow \infty} g(x) = 0$. Then

Ex8 q6

$$\int_a^{\infty} fg < \infty.$$

• (Sum vs Integral test) Suppose $h : [0, \infty) \rightarrow \mathbb{R}^+$ is decreasing then

$$\int_1^{\infty} h \leq \sum_{n=1}^{\infty} h(n) \leq \int_0^{\infty} h.$$

In particular if $\int_0^{\infty} h < \infty$ then $\sum_{n=1}^{\infty} h(n) < \infty$.

WEEK 9,10 - SEQUENCES AND SERIES OF FUNCTIONS

Instead of considering sequences of real numbers we can consider sequences of functions and their convergence. We have two main definition of convergence

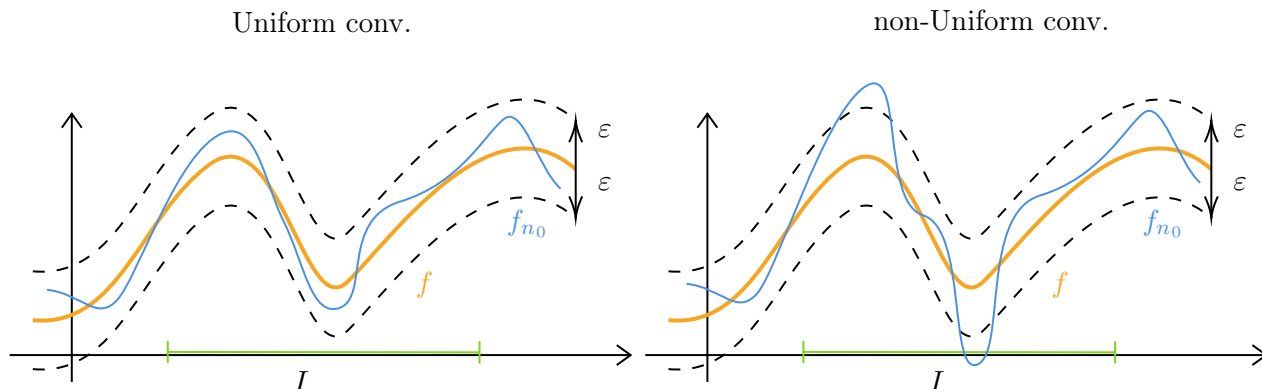
Definition 11. • (Pointwise) $f_n : I \rightarrow \mathbb{R}$ converges pointwise to $f : I \rightarrow \mathbb{R}$ if for all $x_0 \in I$ and all $\varepsilon > 0$ there is $N(\varepsilon, x_0)$ so that for $n \geq N(\varepsilon, x_0)$

$$|f_n(x_0) - f(x_0)| \leq \varepsilon$$

- (Uniform) $f_n : I \rightarrow \mathbb{R}$ converges uniformly to $f : I \rightarrow \mathbb{R}$ if for each $\varepsilon > 0$ there is $N(\varepsilon)$ so that for $n \geq N(\varepsilon)$

$$|f_n(x) - f(x)| \leq \varepsilon \quad \text{for all } x \in I.$$

The only difference between the definitions is the position of the *for all* $x \in I$. In practice this means that pointwise convergence (N depends on x_0 and ε) reduces to the numerical convergence $f_n(x_0) \rightarrow f(x_0)$ whereas with uniform convergence N is independent of x_0 and the inequality holds on the entire interval I . This means that for an arbitrarily thin ε -belt around f all functions f_n for $n \geq N(\varepsilon)$ have to lie within this belt. With $n_0 \geq N$ we have the following pictures:



Exam Tip: To show that f_n does *not* converge uniformly to f : find sequence of points $x_n \in I$ and $\varepsilon > 0$ so that for any n we have

$$|f_n(x_n) - f(x_n)| \geq \varepsilon.$$

Example 4 (x^n convergence on $(0, 1)$): $f_n(x) = x^n$ defined on $(0, 1)$ converges pointwise to $f = 0$. But not uniformly because with $x_n = 2^{-1/n}$ we have $|f_n(x_n) - f(x_n)| = 1/2$ and so we can pick $\varepsilon = 1/3$.

Theorem 20. • Uniform convergence implies pointwise convergence.

- Uniform limit of continuous functions is continuous.
- (Weierstrass M-test) $f_n : I \rightarrow \mathbb{R}$. If for all $x \in I$ $|f_n(x)| \leq M_n$ and $\sum_n M_n < \infty$ then $\sum f_n$ Ex9 q2,3 converges absolutely and uniformly.

Example 5: • (Fourier Series) If $\sum_{-\infty}^{\infty} |a_n| < \infty$ then $\sum_{-\infty}^{\infty} a_n e^{inx}$ converges uniformly on $\mathbb{R}$.

- (Dirichlet Series) If $\sum_1^{\infty} \frac{a_n}{n^s}$ converges for s_0 then it converges for all $s \geq s_0$.

Power Series

Theorem 21. • If $\sum a_n x^n$ converges (pointwise) at x_0 then it converges pointwise on $(-|x_0|, |x_0|)$ and uniformly on any slightly smaller set $(-|x_0| + \varepsilon, |x_0| - \varepsilon)$ with $\varepsilon > 0$.

- (Radius of convergence) The domain of convergence of a power series $\sum a_n x^n$ is $(-R, R)$ and possibly but not necessarily endpoints. Outside this circle the series diverges. ⁵

There is a formula for the radius of convergence.

Ex10 q3

$$R = (\limsup_{n \rightarrow \infty} |a_n|^{1/n})^{-1}$$

⁵Radius because for example in complex analysis the only possible sets on which a power series can converge are actual disks (convergence at the edges has to be separately investigated.)